5.5 Spare part surgery (SUR)

A study of the physics associated with spare part surgery for joint replacements and lens implants:

- mechanical properties of bone and replacement materials
- 'designer' materials for medical uses.

There are opportunities for students to consider ethical issues relating to surgical intervention, and to learn how new scientific knowledge is validated and communicated through peer-reviewed publication.

Students will be assessed on their ability to:		Suggested experiments
22	obtain and draw force-extension, force-compression, and tensile/compressive stress-strain graphs. Identify the *limit* of *proportionality, elastic limit* and *yield point*	Obtain graphs for, for example, copper wire, nylon and rubber
23	investigate and use Hooke's law, $F = k\Delta x$, and know that it applies only to some materials	
24	explain the meaning and use of, and calculate *tensile/compressive stress, tensile/compressive strain, strength, breaking stress, stiffness* and *Young Modulus*. Obtain the Young modulus for a material	Investigations could include, for example, copper and rubber
27	calculate the elastic strain energy E_{el} is a deformed material sample, using the expression $E_{el} = ½F\Delta x$, and from the area under its force/extension graph	